

National AI Action Plan to Advance Artificial Intelligence Leadership in the United States of America

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An AI Action Plan for the United States should be adaptive and continuously updated to ensure sustained leadership in AI. When considering the essential elements required to foster innovation to advance national leadership in AI, one must consider not one, but multiple key components. The basic framework of these components should align with ethical and responsible AI development, strengthening national security, promoting economic growth, and fostering innovation within the AI ecosystem. These components should include:

Research and Development:

- Leveraging partnerships between Universities and Industry will reduce cost and create a competitive playing field for the development of responsible AI development. This collaboration will also reduce the burden on national lab programs, minimizing the necessity for grant funded research and development.

- Centers of excellence should be established in innovation critical sectors such as health care, defense, and education. These centers of excellence would allow for fiscally responsible oversight without stifling innovation.
- Consideration should be made for investment in next generation computing to enhance AI capabilities across these critical sectors and centers of excellence.

Workforce Development and Education:

- The aforementioned centers of excellence provide an environment where public and private partnerships thrive. Artificial Intelligence certification programs and ongoing certification through these partnerships may generate apprenticeships and learning opportunities at a large scale.
- Introducing AI into STEM education at the K-12 level should be a priority in education. Preparing young minds for the advanced AI certification through advanced AI certification programs pathways.
- Additional basic AI literacy training and education should be provided at all levels of education, universities, and workforce.

Ethics Safety and Governance:

- A National Artificial Intelligence Regulatory Framework should be adopted in order to promote fair, accountable, and transparent systems. The National Artificial Intelligence Initiative has developed the “Trustworthy Framework” that address 6 key components of AI governance and safety.
- The regulatory framework should also ensure that human values are considered in AI in order to minimize risk. This would reinforce public trust by reducing discrimination through unintended bias or mode degradation.

National Security and Defense:

- Utilizing AI to defend against adversarial threats with autonomous systems while maintaining human oversight.
- Cybersecurity, threat detection, and mitigation capabilities that are AI powered and have human oversight would enhance national security and reduce the burden on enforcement agencies.

Artificial Intelligence for Economic Growth and Global Competitiveness:

- Provide support for international AI cooperation by utilizing centers of excellence and credentialing to engage in AI global standard setting.
- Setting the standard globally through innovative collaboration between universities and industry creates a fiscally responsible standard that maintains leadership dominance.

Artificial Intelligence Public Sector Innovation:

- Investments in AI driven public services and health care improvements improves and increases efficiency and reduces cost.
- Research into AI driven solutions for energy conservation and disaster response are examples of sustainable AI generated return on investment.

Artificial Intelligence Policy and Public Trust:

- Transparency and explainability in all models used
- Public engagement through academic and industry collaboration to promote education and beneficial AI use cases.
- AI Bill of Rights: Implement a framework protecting individual rights in AI interactions.

Actions to be taken to Achieve National Artificial Intelligence Leadership Objectives and Implement a National AI Action Plan:

1. Research and Development Investment Actions:

- Expand National AI Research Institutes by funding at least 10 new AI research hubs in collaboration with leading universities and private sector partners.
- Increasing NSF and DARPA AI R&D budgets by allocating additional funds to NSF, DARPA, and DOE for fundamental AI research.
- Offer \$500M in innovation prizes through AI Grand Challenges for breakthroughs in quantum AI, Healthcare AI, and autonomous systems.
- Estimated Budget: \$20-25B over 5 years

2. Workforce Development and Education Actions:

- Integrate AI curriculum into K-12 educational programs schools and universities through government grants.
- Establish apprenticeships through centers of excellence and public private partnerships.
- Create AI workforce retaining grants to provide a \$5000 grant for worker transition to AI-related roles.
- Estimated Budget: \$10B over 5 years

3. Establish a National AI Safety and Ethics Office responsible for AI governance and risk assessment.

- Mandate AI impact assessment and return on investment prior to implementing high impact AI systems.

- Develop a regulatory AI Sandbox within government to allow companies to test AI applications under regulatory oversight prior to full scale deployment.

- Budget \$5B over 5 years

4. AI National Security and Defense:

Actions:

- Fund AI driven cybersecurity initiatives and AI powered detection of cyber defense
- Enhance military intelligence applications through autonomous threat analysis with human oversight.
- Launch AI for National Defense Initiative: Establish a secure AI driven cloud infrastructure for defense-related AI infrastructure.

- Estimated Budget: \$25B over 5 years

5. AI economic growth and global competitiveness development

Action:

- Tax incentives for AI start ups.
- Expand AI adoption grants to encourage integration and AI feature development.
- Develop an export strategy to promote U.S. AI products and services globally and attract foreign investment.

- Estimated Budget: \$15B over 5 years

6. AI and Public Sector Innovation Action:

- Deploy AI in federal agencies for fraud detection and benefits processing

- Develop National AI data sharing framework for AI researchers that is secure and ethical
- Launch AI powered energy modeling projects
- Estimated Budget: \$10B over 5 years

7. AI Policy and Public Trust Action:

- Enact AI transparency laws and systems that disclose when AI is used
- Develop public AI literacy campaign
- Create a National Ombudsman Office for AI related complaints and ensure accountability
- Estimated Budget: \$5B over 5 years

Total Estimated Budget: \$90-100B Over 5 years

This budget aligns with the investments made by current AI leaders (China and EU) and ensures that the US remains at the forefront of AI leadership. Funding resources may include:

- Federal Investment Fund
- Public-Private Partnerships
- Reallocation of existing R&D budgets
- AI Specific Taxation (High Revenue AI business

These concrete actions, paired with sustained government commitment and private sector collaboration, will position the U.S. as the world leader in AI innovation.